Submission Summary

Conference Name
29th International Conference on Computer and Information Technology

Paper ID
618

Paper Title
A Multi-Class Machine Learning Approach for Detecting Depression, Anxiety, and Insomnia Risk Levels Based on Gaming Behavioral Patterns

Abstract
Depression, anxiety, and insomnia are among the most common mental health conditions worldwide. Early identification remains difficult because conventional assessment methods rely on surveys and clinical interviews, which require individuals to seek professional help. This paper investigates whether gaming behavior can be used as a behavioral indicator for predicting mental health risk levels. The proposed framework classifies users into Low, Moderate, and High risk categories for depression, anxiety, and insomnia using demographic and gaming-related features collected from 724 participants in Tunisia and Gabon. The dataset includes age, gaming frequency, gaming duration, gamertype scores, device usage, and country. Data preprocessing involved train-test splitting, feature scaling, class balancing using SMOTE, and feature engineering. Seven machine learning models, including Logistic Regression, Random Forest, SVM, KNN, XGBoost, GBM, and MLP, were evaluated together with Voting and Stacking ensemble methods. Stacking achieved the highest accuracy for Depression (80.58%) and Insomnia (84.71%), while XGBoost produced the best result for Anxiety (86.36%). Model performance was further assessed using precision, recall, and five-fold crossvalidation. SHAP and LIME were used to explain the contribution of individual features to the predictions. The findings indicate that gaming behavioral patterns can provide useful information for identifying mental health risk levels and may support future behavioral screening applications.

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Submission Questions Response

1. Paper Template
This paper follows the standard IEEE conference template for writing

Agreement accepted

2. Double-Blind Paper
The author's names, affiliations, postal addresses, and email addresses are removed from the paper during initial submission. (Please note, it's a double-blind review. Authors are requested to remove their names from initial submission.)

Agreement accepted

3. Paper length
Submitted paper is limited to 4-6 pages in IEEE 2-column format, including all figures, tables, equations, and references

Agreement accepted

4. Plagiarism
The similarity index of this paper with references is less than 30%

Agreement accepted

5. Would you, or any of your co-authors with a PhD, be interested in serving as a reviewer?
If you are willing, please provide the email addresses of yourself or any co-authors with a PhD who would be interested in serving as reviewers. Kindly list their names and emails below, separated by commas.

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